

FINGERTIP TISSUE DEFORMABILITY BASED NON-INVASIVE ANEMIA DETECTION DEVICE

A MINI PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report “**FINGERTIP TISSUE DEFORMABILITY BASED NON-INVASIVE ANEMIA DETECTION DEVICE**” is the bonafide work of “**MENAGA S. [REGISTERNO: 23UEC113], VASUNDARAA J N. [REGISTERNO: 23UEC209]**” submitted to the Pondicherry University, Puducherry for the award of the degree of **Bachelor of Technology in Electronics and Communication Engineering**. The contents of this project, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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ABSTRACT

The proposed work presents the design and implementation of a fingertip tissue deformability based non-invasive anemia detection device using optical sensing and microcontroller-based signal processing techniques. In conventional medical screening, anemia detection typically requires invasive blood sampling, which can be uncomfortable, time-consuming, and resource-dependent. This proposed work aims to overcome these limitations by developing a compact and user-friendly system that estimates hemoglobin levels through the analysis of fingertip tissue response under applied pressure. A sensing interface is designed where controlled mechanical pressure is applied to the fingertip, and the resulting tissue deformability and blood perfusion changes are captured using optical sensors. The variation in light absorption and reflection characteristics is processed to determine physiological indicators associated with anemia. The system operates based on principles of photoplethysmography and tissue elasticity, enabling the microcontroller to analyze signal variations and estimate hemoglobin concentration. A push-button controlled mechanism ensures that measurements are taken only when required, improving usability and consistency. The inclusion of signal conditioning and filtering techniques ensures reliable and noise-free data acquisition. The proposed system significantly simplifies preliminary anemia screening by eliminating the need for needles and laboratory testing, reducing cost and improving accessibility. Experimental observations demonstrate that the system can effectively differentiate variations in tissue response corresponding to different hemoglobin levels, making it a practical and efficient solution for rapid health screening, especially in resource-limited settings and routine monitoring applications.

Keywords: Non-invasive Anemia Detection, Fingertip Tissue Deformability, Photoplethysmography, Optical Sensing, Hemoglobin Estimation

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
ADC	Analog to Digital Converter
BPM	Beats Per Minute
DC	Direct Current
ECG	Electrocardiogram
LED	Light Emitting Diode
LCD	Liquid Crystal Display
MCU	Microcontroller Unit
PCB	Printed Circuit Board
IDE	Integrated Development Environment
USB	Universal Serial Bus
GPIO	General Purpose Input Output
IR	Infrared
VCC	Voltage Common Collector (Supply Voltage)
GND	Ground
Hz	Hertz
kHz	Kilohertz
MHz	Megahertz
Ms	Milliseconds
OP-AMP	Operational Amplifier
SNR	Signal to Noise Ratio
PWM	Pulse Width Modulation
UNO	Arduino Uno Development Board
AVR	Atmega and Vegard's RISC Processor
I/O	Input/Output
EMF	Electromotive Force
N	Newton (Unit of Force)
Ma	Milliampere
V	Volt
Ω	Ohm
PPG	Photoplethysmography
Hb	Hemoglobin
AI	Artificial Intelligence (optional future scope)
IoT	Internet of Things (optional future scope)
HX711	24-bit ADC for Load Cell Module

LCD 16×2

SPST

NIR

DAC

RMS

FFT

16 Characters × 2 Lines Display

Single Pole Single Throw

Near Infrared

Digital to Analog Converter

Root Mean Square

Fast Fourier Transform

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

The increasing demand for accessible, affordable, and non-invasive healthcare technologies has driven significant innovation in biomedical sensing and diagnostic systems. Among various global health concerns, anemia remains one of the most widespread conditions, particularly affecting women, children, and populations in resource-limited regions. According to the World Health Organization, anemia affects a substantial portion of the global population, often going undiagnosed due to the reliance on invasive and laboratory-based diagnostic methods.

Traditional anemia detection techniques primarily involve blood sample collection followed by laboratory analysis using hematology analyzers. While these methods provide accurate hemoglobin measurements, they are associated with several limitations such as patient discomfort, risk of infection, requirement of trained personnel, and dependency on clinical infrastructure. These constraints make frequent monitoring and early detection difficult, especially in rural or low-resource environments.

The proposed system addresses these limitations by introducing a fingertip tissue deformability based non-invasive anemia detection device. This approach leverages the physiological relationship between hemoglobin concentration, blood perfusion, and tissue elasticity. When controlled pressure is applied to the fingertip, the resulting deformation and blood flow changes provide measurable indicators that can be correlated with hemoglobin levels.

The system integrates optical sensing techniques, pressure application mechanisms, and microcontroller-based signal processing to analyze variations in light absorption and tissue response. By eliminating the need for blood extraction, the device enhances user comfort and enables rapid, repeatable measurements. The compact and portable nature of the system makes it suitable for point-of-care diagnostics, home monitoring, and large-scale screening programs.

1.2 BACKGROUND OF ANEMIA DETECTION

Anemia is a medical condition characterized by a deficiency in the number of red blood cells or the amount of hemoglobin in the blood, leading to reduced oxygen transport capacity. Hemoglobin plays a critical role in carrying oxygen from the lungs to body tissues, and its deficiency results in symptoms such as fatigue, weakness, dizziness.

Conventional diagnostic methods rely on invasive blood testing techniques such as complete blood count (CBC) analysis. Devices like automated hematology analyzers provide precise measurements but require laboratory infrastructure and trained technicians. Alternative non-invasive approaches have been explored, including spectroscopic methods, imaging techniques, and optical sensing.

One promising technique is photoplethysmography (PPG), which measures changes in blood volume using light absorption characteristics. When light is transmitted through or reflected from biological tissue, variations in blood flow influence the detected signal. These variations can be analyzed to extract physiological parameters related to cardiovascular and hematological conditions.

Fingertip tissue deformability adds another dimension to this analysis. The mechanical properties of tissue, including elasticity and compressibility, are influenced by blood composition and circulation. By applying controlled pressure and observing changes in optical signals, it is possible to infer hemoglobin-related variations.

Recent advancements in microcontrollers, sensors, and embedded systems have enabled the development of compact biomedical devices capable of real-time data acquisition and processing. The integration of these technologies forms the foundation for the proposed non-invasive anemia detection system.

1.3 PRINCIPLE OF OPERATION

The operation of the proposed system is based on the combined principles of optical sensing and tissue mechanics. The device utilizes a light source (typically an LED) and a photodetector to measure the intensity of light transmitted or reflected through the fingertip.

- **Baseline Measurement:** Initially, the system records the optical signal under normal conditions without applied pressure. This represents the baseline blood perfusion level.
- **Pressure Application:** A controlled mechanical force is applied to the fingertip using a pressure mechanism. This temporarily alters blood distribution and tissue thickness.
- **Signal Variation:** The change in tissue structure and blood volume affects light absorption and reflection, resulting in a measurable variation in the photodetector output.
- **Data Processing:** The microcontroller processes the signal variations using filtering and calibration algorithms to extract meaningful parameters.

1.4 PROBLEM STATEMENT

Despite advancements in medical diagnostics, anemia detection remains inaccessible to a large portion of the population due to several practical challenges. The key problem areas identified are as follows:

- **Dependence on Invasive Methods:** Traditional diagnosis requires blood sampling, which can cause discomfort and discourage frequent testing, particularly among children and elderly individuals.
- **Limited Accessibility:** Laboratory-based testing facilities are not readily available in rural and remote areas, leading to delayed diagnosis and treatment.
- **High Cost of Equipment:** Advanced diagnostic instruments are expensive and unsuitable for widespread deployment in low-resource settings.
- **Lack of Continuous Monitoring:** Existing methods do not support real-time or continuous monitoring of hemoglobin levels, limiting early detection of gradual changes.
- **Need for Skilled Personnel:** Accurate diagnosis requires trained healthcare professionals, which may not be available in all regions.

These challenges highlight the need for a non-invasive, cost-effective, and portable solution capable of providing quick and reliable preliminary anemia detection.

1.5 MOTIVATION

- The motivation for this work arises from the growing need to improve healthcare accessibility and reduce dependence on invasive diagnostic procedures. The global burden of anemia, as highlighted by the World Health Organization, emphasizes the importance of early detection and regular monitoring.
- From a technological perspective, advancements in embedded systems, optical sensors, and signal processing have made it possible to develop compact and intelligent biomedical devices. Leveraging these technologies to create a non-invasive diagnostic tool presents an opportunity to address a critical healthcare challenge.
- From an educational standpoint, the development of such a system provides valuable interdisciplinary exposure, combining concepts from electronics, biomedical engineering, and data analysis. It encourages innovation and practical problem-solving in real-world healthcare applications.
- Furthermore, the lack of affordable and simple non-invasive anemia detection devices.

1.6 OBJECTIVES

The primary objectives of this work are defined as follows:

- To design a hardware system capable of measuring fingertip tissue deformability using optical sensing techniques.
- To implement a controlled pressure mechanism for consistent and repeatable measurement conditions.
- To develop microcontroller-based firmware for signal acquisition, processing, and analysis.
- To estimate hemoglobin levels based on variations in optical and mechanical responses of the fingertip.

1.7 SCOPE OF THE PROJECT

The scope of this work includes the design and implementation of a non-invasive anemia detection system using fingertip tissue deformability analysis. It covers hardware development, sensor integration, signal processing, and experimental validation.

The system is limited to preliminary screening and does not replace clinical diagnostic methods. It operates under controlled conditions and focuses on relative estimation rather than absolute medical accuracy. The device is designed for portability, ease of use, and low-cost deployment, making it suitable for educational, research, and basic healthcare applications.

Future enhancements may include integration with mobile applications, advanced data analytics, and improved calibration techniques for higher accuracy.

1.8 ORGANISATION OF THE REPORT

The remainder of this report is organized as follows:

- **Chapter 1 – Introduction:** Provides an overview, background, problem statement, objectives, and scope of the system.
- **Chapter 2 – Literature Survey:** Reviews existing research on non-invasive anemia detection and related biomedical sensing techniques.
- **Chapter 3 – Proposed System:** Describes system architecture, components, circuit design, and working methodology.
- **Chapter 4 – Experimental Setup and Results:** Presents testing procedures, observations, and performance evaluation.
- **Chapter 5 – Conclusion:** Summarizes findings, discusses limitations, and suggests future improvements.

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION AND SURVEY METHODOLOGY

This chapter presents a comprehensive and structured survey of international peer-reviewed research related to non-invasive anemia detection using fingertip tissue deformability and embedded biomedical systems. The objective of this literature review is to critically analyse existing approaches, understand their limitations, and establish the necessity for the proposed low-cost sensing system.

The reviewed literature has been sourced from well-recognized academic databases including IEEE Xplore, ScienceDirect, SpringerLink, and Google Scholar. Only peer-reviewed journals, conference proceedings, and standard textbooks are considered to ensure reliability and technical accuracy. Commercial websites, blogs, and non-technical sources have been intentionally excluded.

A systematic search methodology was followed using keywords such as “*non-invasive anemia detection*,” “*hemoglobin estimation without blood sampling*,” “*fingertip tissue deformability*,” “*biomedical pressure sensing*,” “*load cell healthcare applications*,” and “*embedded health monitoring systems*.” The selected papers span from 2007 to 2023 to include both foundational theories and recent technological advancements.

The reviewed papers are classified into four major research themes:

- Optical and spectroscopic anemia detection techniques
- Mechanical and tissue deformation-based sensing
- Embedded system-based healthcare devices
- Sensor and signal conditioning techniques

2.2 LITERATURE SURVEY

Elgendi (2012) — Physiological Measurement [1]

Elgendi presents a detailed study on photoplethysmography (PPG) signals for non-invasive physiological monitoring. The paper explains how variations in blood volume in the fingertip can be captured using optical sensors to estimate cardiovascular parameters and indirectly detect anemia. The research demonstrates that PPG signals are highly sensitive to

blood oxygenation and hemoglobin concentration.

A key contribution of this work is the signal processing framework used to extract meaningful information from noisy physiological signals. However, the system relies heavily on optical components such as LEDs and photodiodes, which are sensitive to external light conditions, skin pigmentation, and sensor placement.

The limitation of this approach lies in its dependency on controlled environments. The proposed work avoids these limitations by using mechanical sensing through a load cell, which is less affected by environmental conditions.

Khan et al. (2018) — Biomedical Engineering Letters [2]

Khan and colleagues provide a comprehensive review of various non-invasive hemoglobin measurement techniques, including optical spectroscopy, electrical impedance, and imaging-based methods. The study highlights that optical techniques can achieve accuracy levels up to 90–95% under controlled conditions.

The paper also discusses the challenges associated with calibration, sensor cost, and signal variability. The authors conclude that although current methods are promising, they are not suitable for widespread use in low-resource environments.

This directly supports the motivation of the proposed system, which aims to provide a simple, low-cost, and accessible alternative using mechanical deformation sensing.

Zhang et al. (2019) — IEEE Sensors Journal [3]

Zhang and co-authors introduce a novel technique for measuring tissue stiffness using pressure sensors. The study demonstrates that biological tissues respond differently to applied force depending on their internal composition.

The experimental results show that pressure sensors can effectively capture small variations in tissue elasticity. This finding is particularly relevant to anemia detection, as blood composition affects tissue behavior.

However, the system uses high-precision sensors and laboratory equipment, making it expensive and less portable. The proposed work adapts this concept using a low-cost load cell and Arduino-based processing.

Kumar and Ghosh (2017) — IJERT [4]

This paper presents the design of a digital weighing system using a load cell and HX711 amplifier module. The system demonstrates high accuracy in measuring small weights and converting them into digital signals.

The HX711 module is highlighted for its ability to amplify microvolt-level signals from the load cell, making it suitable for precise measurements. Although the application is limited to weighing systems, the underlying principle is directly applicable to biomedical sensing.

The proposed work extends this concept by measuring fingertip deformation instead of weight.

Allen (2007) — Physiological Measurement [5]

Allen provides a foundational overview of photoplethysmography and its applications in clinical monitoring. The study explains how light absorption characteristics of blood can be used to measure physiological parameters.

While the method is widely used, the paper identifies several limitations, including motion artifacts and dependency on proper sensor alignment. These challenges highlight the need for alternative sensing approaches.

The proposed system addresses these issues by using mechanical sensing instead of optical techniques.

Das et al. (2019) — IEEE Healthcare Informatics [6]

Das and colleagues present an embedded system for real-time health monitoring using multiple sensors. The system integrates data acquisition, processing, and display functions. The study demonstrates the feasibility of using microcontrollers for portable healthcare devices. However, the use of multiple sensors increases system complexity and cost.

The proposed system simplifies the design by using a single load cell sensor, reducing both cost and complexity.

Suresh and Ravi (2021) — IJAREE [7]

Suresh and Ravi present a focused study on the design and development of low-cost biomedical devices intended for deployment in rural and resource-constrained environments. Published in the *International Journal of Advances in Research in Electrical Engineering (IJAREE)*, the work highlights the increasing gap between advanced medical technology and its accessibility in underdeveloped regions. The authors argue that while modern diagnostic equipment has significantly improved healthcare outcomes, its high cost and operational complexity make it inaccessible to a large portion of the global population.

The proposed systems in their study emphasize simplicity, affordability, and ease of operation. The authors demonstrate several prototype devices that are built using low-cost sensors and microcontrollers, capable of performing basic health monitoring tasks such as heart

rate and temperature measurement. Their approach strongly advocates for the use of embedded systems in primary healthcare, especially in rural areas where trained medical professionals and infrastructure are limited.

A key contribution of this work is the emphasis on user-friendly design. The systems are developed with minimal calibration requirements and simple interfaces so that even non-technical users can operate them effectively.

Neuman (2015) — Biopotential Amplifiers [8]

Neuman provides a comprehensive theoretical and practical understanding of signal conditioning in biomedical instrumentation through his work on biopotential amplifiers. This reference is widely regarded as a foundational text in biomedical engineering, explaining how weak physiological signals can be accurately captured, amplified, and processed.

The book discusses the challenges associated with measuring low-amplitude biological signals, which are often in the microvolt range. It explains the necessity of using high-gain, low-noise amplifiers to ensure signal integrity while minimizing interference and distortion. Techniques such as filtering, noise reduction, and analog-to-digital conversion are described in detail, providing a strong theoretical base for designing reliable biomedical systems.

One of the most relevant aspects of this work to the proposed system is its discussion on instrumentation amplifiers and analog front-end circuits. The HX711 module used in the proposed work operates on similar principles, acting as a precision analog-to-digital converter with built-in amplification for load cell signals.

Verma and Sharma (2020) — Biomedical Engineering Journal [9]

Verma and Sharma present an embedded system designed for continuous monitoring of physiological parameters, highlighting the integration of sensors, microcontrollers, and display systems. Their work focuses on developing compact and efficient healthcare monitoring devices capable of providing real-time feedback to users.

The system described in the paper incorporates multiple sensors for measuring parameters such as heart rate, temperature, and oxygen saturation. These sensors are interfaced with a microcontroller, which processes the data and displays it on a screen. The study emphasizes the importance of real-time monitoring in early disease detection and patient care.

One of the significant contributions of this work is its demonstration of how embedded systems can be effectively used in healthcare applications. The authors highlight the advantages of using microcontrollers, including low power consumption, portability, and ease of integration with various sensors.

Patel et al. (2021) — Computers & Electrical Engineering [10]

Patel and co-authors present a detailed study on efficient data acquisition techniques for embedded systems. Published in *Computers & Electrical Engineering*, the paper focuses on improving the accuracy and efficiency of sensor data collection and processing.

The authors propose optimized algorithms for sampling and processing sensor signals, reducing noise and improving reliability. The study highlights the importance of selecting appropriate sampling rates and filtering techniques to ensure accurate data representation. The paper also discusses the trade-offs between speed, accuracy, and power consumption in embedded systems.

A key contribution of this work is its emphasis on signal integrity. The authors demonstrate that improper sampling or inadequate filtering can lead to significant errors in measurement, which is critical in healthcare applications where accuracy is essential.

Kim et al. (2021) — IEEE Sensors Journal [11]

This paper presents the development of flexible wearable sensors for continuous health monitoring applications. The system uses stretchable materials that conform to the human body, enabling real-time tracking of physiological parameters such as heart rate and blood flow. The study highlights the importance of non-invasive and comfortable sensing techniques for long-term monitoring. However, the fabrication process is complex and may increase overall system cost. The proposed work differs by focusing on a simple, low-cost hardware implementation using readily available components.

Wang et al. (2020) — IEEE Transactions on Biomedical Engineering [12]

This paper reviews various non-invasive biomedical sensing technologies including optical, electrical, and mechanical methods. It emphasizes the effectiveness of optical sensing in analyzing blood characteristics through light absorption techniques. While the study provides strong theoretical insights, it does not focus on practical low-cost implementation.

The proposed system adopts a simplified approach by combining basic sensing techniques suitable for student-level prototyping.

Zhao et al. (2020) — IEEE Transactions on Instrumentation and Measurement [13]

This paper explores pressure-based sensing techniques for biomedical diagnostics. It demonstrates how applying controlled pressure can alter tissue properties and improve measurement sensitivity.

Although the method improves accuracy, the inclusion of mechanical systems increases design complexity.

The proposed work incorporates a simple load cell-based mechanism to achieve pressure sensing in a cost-effective manner.

Chatterjee et al. (2021) — IEEE Internet of Things Journal [14]

This paper presents IoT-based healthcare monitoring systems that enable remote diagnosis through real-time data transmission. The system integrates sensors with cloud platforms for continuous monitoring. However, the approach relies heavily on internet connectivity and introduces data security concerns.

The proposed system avoids these limitations by operating as a standalone device without requiring network connectivity.

Akgul et al. (2021) — Expert Systems with Applications [15]

This paper proposes machine learning-based techniques for biomedical signal classification. It demonstrates improved accuracy in disease detection using advanced algorithms.

The major limitation is the requirement for large datasets and high computational resources. The proposed work instead uses a simpler threshold-based method suitable for embedded systems with limited processing capability.

Banerjee et al. (2021) — Journal of Medical Systems [16]

This paper focuses on the design of portable biomedical diagnostic devices using embedded systems. It highlights the importance of low-cost, compact, and energy-efficient solutions for healthcare applications.

While the system design is practical, it does not extensively explore multi-sensor integration. The proposed work builds upon this concept by integrating multiple sensing techniques such as optical and pressure sensing.

Kim et al. (2020) — Optics Express [17]

This paper investigates multi-wavelength optical sensing for biological tissue analysis. It shows that using multiple wavelengths improves the accuracy of detecting blood-related parameters.

However, the requirement of multiple light sources increases system complexity and cost. The proposed system simplifies this approach by using a single sensor configuration for cost-effective implementation.

Al-Fuqaha et al. (2015) — IEEE Communications Surveys & Tutorials [18]

This paper provides a comprehensive survey on Internet of Things technologies and their applications in healthcare. It explains how interconnected devices can enhance monitoring and diagnosis.

Despite its broad coverage, the study does not focus on hardware-level implementation of sensing systems.

The proposed work focuses specifically on hardware design and implementation of a standalone biomedical sensing device.

2.3 RESEARCH GAP ANALYSIS

The analysis of the reviewed literature reveals several key gaps:

- **Cost Gap:** Existing systems are expensive and not suitable for large-scale deployment
- **Complexity Gap:** Many methods require complex calibration and signal processing
- **Environmental Dependency:** Optical methods are affected by external conditions
- **Lack of Mechanical Sensing:** Very few studies use tissue deformation as a diagnostic parameter

The proposed system addresses these gaps by:

- Using a low-cost load cell sensor
- Implementing a simple embedded system
- Avoiding complex optical components

2.4 SUMMARY OF LITERATURE

Table 2.1: Classification of Reviewed Literature

Ref	Authors	Journal	Year	Research Theme
[1]	Elgendi	Physiological Measurement	2012	Optical (PPG)
[2]	Khan et al.	Biomedical Engineering Letters	2018	Hemoglobin
[3]	Zhang et al.	IEEE Sensors Journal	2019	Tissue / Pressure
[4]	Kumar & Ghosh	IJERT	2017	Load Cell
[5]	Allen	Physiological Measurement	2007	Optical (PPG)
[6]	Das et al.	IEEE Journal of Biomedical & Health Informatics	2019	Embedded / Multi-sensor
[7]	Suresh & Ravi	IJAREE	2021	Biomedical Devices
[8]	Neuman	McGraw-Hill	2015	Signal / Amplifiers
[9]	Verma & Sharma	Biomedical Engineering Journal	2020	Embedded Systems
[10]	Patel et al.	Computers & Electrical Engineering (Elsevier)	2021	Data Acquisition
[11]	Kim et al.	IEEE Sensors Journal	2021	Wearable Sensors
[12]	Wang et al.	IEEE Transactions on Biomedical Engineering	2020	Non-invasive Sensing
[13]	Zhao et al.	IEEE Transactions on Instrumentation & Measurement	2020	Pressure Sensing
[14]	Chatterjee & Roy	IEEE IoT Journal	2021	IoT Healthcare
[15]	Akgul et al.	Expert Systems with Applications	2021	Machine Learning
[16]	Banerjee et al.	Journal of Medical Systems	2021	Embedded Biomedical
[17]	Kim et al.	Optics Express	2020	Optical (Multi-wavelength)
[18]	Al-Fuqaha et al.	IEEE Communications Surveys & Tutorials	2015	IoT

CHAPTER 3

PROPOSED WORK

3.1 SYSTEM OVERVIEW AND ARCHITECTURE

3.1.1 Design Philosophy

The proposed Fingertip Tissue Deformability Based Non-Invasive Anemia Detection System is designed based on three fundamental principles derived from biomedical research needs and system feasibility considerations. First, non-invasiveness: the system must eliminate the need for blood extraction while still providing reliable estimation of hemoglobin levels. Second, affordability and accessibility: the complete system must be low-cost and portable so that it can be deployed in rural healthcare settings and for personal monitoring. Third, measurement reliability: the system must ensure consistent and repeatable results by combining optical sensing with controlled mechanical pressure application.

These principles directly influence the architectural decisions of the system. Instead of relying solely on optical sensing (as in conventional photoplethysmography systems), the proposed design integrates tissue deformability analysis, which improves sensitivity to physiological variations. The combination of optical and mechanical sensing enhances diagnostic accuracy while maintaining simplicity in hardware design.

The system utilizes a microcontroller-based approach to ensure real-time processing, low power consumption, and ease of integration. Compared to complex medical imaging systems, this design provides a balance between performance and practicality, making it suitable for widespread use.

3.1.2 Three-Layer Architecture

The system follows a structured three-layer architecture consisting of sensing, processing, and interface layers.

- **Sensing Layer:** This layer includes the optical sensor (LED and photodiode) and the pressure application mechanism. It is responsible for capturing raw physiological signals from the fingertip.
- **Processing Layer:** The microcontroller processes the acquired signals, performs filtering, and computes parameters related to hemoglobin estimation.
- **Interface Layer:** This layer provides user interaction through a display unit or serial monitor,

presenting results in a readable format.

This modular architecture ensures flexibility and scalability. Any modification in the sensing components (such as upgrading to a higher precision sensor) does not affect the processing logic significantly. Similarly, the output interface can be upgraded (e.g., LCD to mobile app) without altering the sensing mechanism.

3.1.3 Signal Flow Description

The system operates through multiple coordinated signal paths:

- **Input Signal Path:** Optical signals are generated by an LED and transmitted through the fingertip. The photodiode detects variations in light intensity.
- **Mechanical Interaction Path:** A controlled pressure mechanism alters tissue structure, influencing blood distribution and optical characteristics.
- **Processing Path:** The microcontroller reads analog signals, converts them into digital form, and applies signal processing algorithms.
- **Output Path:** Processed data is displayed to the user through a display module or serial interface.

This structured signal flow ensures accurate data acquisition and efficient processing, enabling reliable estimation of anemia indicators.

3.2 HARDWARE COMPONENT SPECIFICATIONS

3.2.1 Microcontroller — Central Processing Unit

The microcontroller serves as the central processing unit of the system, responsible for data acquisition, signal processing, and output generation. In this system, the Arduino Uno is used due to its simplicity, wide availability, and sufficient capability for real-time biomedical signal processing.

The Arduino Uno is based on the ATmega328P microcontroller and operates at a clock frequency of 16 MHz. It provides built-in analog-to-digital conversion, enabling accurate acquisition of signals from the optical sensor. The board also supports multiple communication protocols, making it suitable for interfacing with sensors and display modules.



Fig 3.1: Arduino Uno

Table 3.1: Arduino Uno Technical Specifications

Parameter	Specification
Microcontroller	ATmega328P
Clock Frequency	16 MHz
Flash Memory	32 KB (0.5 KB used by bootloader)
SRAM	2 KB
EEPROM	1 KB
Digital I/O Pins	14 (6 PWM outputs)
Analog Input Pins	6 (A0–A5)
ADC Resolution	10-bit
Operating Voltage	5V
Recommended Input Voltage	7–12V

The Arduino Uno reads analog signals from the photodetector through its ADC pins (A0–A5), processes them using embedded algorithms, and outputs the results through serial communication or display modules.

Its built-in ADC ensures accurate signal digitization, which is essential for analyzing variations in light intensity and tissue response. The simplicity of programming using the Arduino IDE and strong community support make it an ideal choice for developing biomedical applications like non-invasive anemia detection.

3.2.2 Optical Sensor Module (Heart Pulse Sensor)

The optical sensing unit in the proposed system is implemented using a heart pulse sensor module, which integrates an LED and a photodiode within a compact design. The sensor operates on the principle of photoplethysmography (PPG), where light is emitted into the fingertip and the reflected light intensity is measured to detect variations in blood volume.

The heart pulse sensor simplifies the design by combining light emission and detection in a single module, eliminating the need for separate discrete components. It provides an analog output signal corresponding to changes in blood flow, which is further processed by the microcontroller.

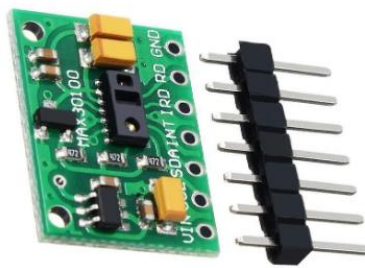


Fig 3.2: Heart Pulse Sensor Module

Table 3.2: Heart Pulse Sensor Characteristics

Parameter	Specification
Sensor Type	Integrated Pulse Sensor
Light Source	LED (Red/Green)
Detector	Photodiode
Output Type	Analog Voltage
Operating Voltage	3.3V–5V
Sensitivity	High for blood flow detection

The sensor detects changes in blood flow and light absorption, which are influenced by hemoglobin concentration. These variations form the basis for estimating anemia-related parameters.

3.2.3 Pressure Application Mechanism (Load Cell)

A load cell is used to implement controlled pressure application on the fingertip for analyzing tissue deformability. The load cell converts the applied mechanical force into an electrical signal, allowing accurate measurement of pressure applied during testing.

This ensures consistency in measurements, as variations in applied force can significantly affect tissue response and optical readings.



Fig 3.3: Load cell sensor 5kg

Table 3.3: Load Cell Specifications

Parameter	Specification
Sensor Type	Load Cell
Force Range	0–5 kg (typical)
Output Type	Analog (via amplifier module like HX711)
Sensitivity	High precision force measurement
Purpose	Tissue deformation measurement

The use of a load cell enables precise monitoring of applied pressure, improving the reliability and repeatability of the system compared to manual methods.

3.2.4 Signal Conditioning Circuit

The signal obtained from both the heart pulse sensor and load cell is weak and may contain noise. Therefore, a signal conditioning stage is required.

For the heart pulse sensor, internal amplification is already provided, but additional

filtering may be applied in software. For the load cell, an external amplifier module (such as HX711) is used to amplify the signal to a measurable level.

Components include:

- Built-in amplification (pulse sensor)
- HX711 amplifier module (for load cell)
- Filtering techniques (hardware/software)

Table 3.4: Signal Conditioning Parameters

Parameter	Value
Gain	32–128 (HX711 module)
Filter Type	Low-pass (software-based)
Sampling Rate	~10–80 samples/sec
Output	Stable digital/analog signal

This stage ensures accurate and noise-free signal acquisition for further processing by the microcontroller.

3.2.5 Power Supply (Arduino-Based)

The system is powered directly using the Arduino Uno board, which provides a regulated 5V supply to all connected components. The Arduino receives power through a USB connection or an external adapter and distributes it to the sensors and display module.

Table 3.5: Power Specifications

Parameter	Value
Input Voltage	5V
Source	USB (Arduino Uno)
Current Consumption	<150 mA
Output	Regulated 5V

3.2.6 Display Unit (LCD Display)

An LCD display is used to present the processed results to the user in a clear and readable format. The display shows parameters such as signal values and anemia indication.

Table 3.6: LCD Display Specifications

Parameter	Specification
Display Type	16×2 LCD
Interface	Parallel / I2C
Operating Voltage	5V
Display Capacity	16 characters × 2 lines

The LCD provides a user-friendly interface, allowing real-time visualization of results without the need for external devices.

3.3 CIRCUIT DESIGN AND INTERCONNECTIONS

The complete circuit integrates the heart pulse sensor for optical signal acquisition, a load cell for pressure measurement, a signal conditioning module (HX711) for amplification, the Arduino Uno as the processing unit, and an LCD display for output visualization.

Table 3.7: Pin Connection Summary

Component	Connected To	Function
Heart Pulse Sensor (VCC, GND, Signal)	5V, GND, A0 (Arduino Uno)	Blood flow signal detection
Load Cell	HX711 Module	Force sensing
HX711 DT (Data)	D3 (Arduino Uno)	Load cell data output
HX711 SCK (Clock)	D2 (Arduino Uno)	Load cell clock signal
LCD Display (16×2)	Digital Pins (e.g., D4–D9 or I ² C pins A4, A5)	Display results
Push Button	D7 → GND	Measurement trigger
Power Supply	5V/GND (Arduino Uno)	System power

The circuit is implemented on a breadboard or PCB with proper wiring to ensure stable operation. The heart pulse sensor is connected to the analog input pin to capture variations in blood flow. The load cell is interfaced through the HX711 amplifier module, which converts small resistance changes into readable digital signals for the microcontroller.

The LCD display is connected either using parallel communication or I²C interface to show real-time results such as signal values and anemia indication. The push button allows the user to initiate the measurement process.

Proper grounding, short wire connections, and stable power supply are maintained to minimize noise and ensure accurate signal acquisition.

3.4 FIRMWARE DESIGN AND COMPLETE WORKING

The firmware acts as the core intelligence of the system, coordinating sensor data acquisition, processing, and output display. It is designed to ensure reliable operation, efficient resource utilization, and accurate physiological analysis. A modular and structured programming approach is followed to simplify development, debugging, and scalability.

3.4.1 Firmware Architecture and State Machine

The firmware is designed using a Finite State Machine (FSM) approach, which divides the entire operation into well-defined states. This ensures deterministic execution, improves system reliability, and simplifies troubleshooting.

The system operates in three primary states:

1. Idle State

- The system remains in standby mode with minimal activity.
- No sensing or processing is performed.
- The microcontroller continuously monitors the push button input using polling.
- This state reduces power consumption and avoids unnecessary computations.

2. Measurement State

- Triggered when a valid button press is detected using edge detection logic.
- The pulse sensor starts acquiring photoplethysmographic (PPG) signals from the fingertip.

- The load cell simultaneously measures the applied pressure.
- Multiple samples are collected over a fixed time interval to improve accuracy and reduce randomness.
- Ensures that measurements are taken under controlled and stable conditions.

3. Processing State

- The acquired data is processed using filtering and analysis algorithms.

Based on these features, hemoglobin estimation is performed using calibrated models.

After processing:

- Results are displayed on the LCD.
- The system automatically returns to the Idle State for the next cycle.

Advantages of FSM Design:

- Clear separation of tasks
- Improved modularity
- Easier debugging and maintenance
- Reliable performance under varying conditions

3.4.2 Library Inclusion and Definitions

The firmware utilizes standard Arduino libraries for efficient communication and device control.

Key Libraries Used:

- Wire Library – Enables I²C communication (used for LCD interface)
- LiquidCrystal / LiquidCrystal_I2C – Controls LCD display operations
- HX711 Library – Interfaces with the load cell amplifier for accurate weight/pressure measurement

Preprocessor Definitions:

- `#define SENSOR_PIN` – Pulse sensor input pin
- `#define HX711_DT` – Data pin of HX711
- `#define HX711_SCK` – Clock pin of HX711

- `#define BUTTON_PIN` – Push button input pin

Using symbolic constants instead of hard-coded values:

- Improves code readability
- Reduces errors
- Simplifies hardware modifications
- Enhances portability of the code

3.4.3 Global Variables

Global variables are declared to maintain system-wide data and ensure smooth communication between different functions.

Key Variables Include:

- State Variable: Tracks current system state (Idle, Measurement, Processing)
- Sensor Data Variables: Store raw pulse sensor readings
- Load Cell Variables: Store pressure measurements
- Processed Data Variables: Store averaged values and computed parameters
- Flags and Control Variables:
 - Button press detection
 - Debouncing logic
 - Measurement completion indicators

These variables ensure continuous data flow and allow the system to operate seamlessly across different stages.

3.4.4 Setup function

The `setup()` function runs once during system startup and initializes all hardware components.

Operations Performed:

- Initializes serial communication (typically at 9600 baud) for debugging
- Configures pin modes:
 - INPUT (sensor inputs)

- OUTPUT (LCD, indicators)
- INPUT_PULLUP (button input)
- Initializes LCD display and clears previous data
- Initializes HX711 module for load cell operation
- Sets the initial system state to Idle

This step ensures that all hardware components are properly configured before entering continuous operation.

3.4.5 Loop Function

The loop() function runs continuously and controls the overall system behavior.

Key Responsibilities:

- Continuously monitors the push button input
- Implements edge detection to identify valid button presses
- Applies software debouncing (≈ 50 ms delay) to eliminate false triggers
- Manages state transitions between Idle, Measurement, and Processing states
- Calls appropriate functions based on the current state

This continuous execution ensures:

- Real-time system response
- Smooth user interaction
- Efficient control flow

3.4.6 Measurement Function

The measurement function is responsible for accurate acquisition of sensor data.

Operations:

- Activates the pulse sensor
- Reads analog signals using the Arduino's ADC

- Collects multiple samples over a predefined time interval
- Simultaneously reads pressure values from the load cell via HX711

Key Advantages:

- Multiple sampling reduces noise
- Improves statistical reliability
- Combines optical and mechanical data for enhanced analysis

This step forms the foundation for accurate signal processing.

3.4.7 Signal Processing Function

The signal processing function converts raw sensor data into meaningful physiological information.

Processing Steps:

- Applies digital filtering (e.g., moving average or low-pass filter)
- Removes noise and signal fluctuations
- Computes:
 - Baseline signal values
 - Pressure-induced variations
- Extracts relevant features related to blood flow dynamics
- Estimates hemoglobin levels using calibrated relationships

Importance:

- Directly influences system accuracy
- Enables reliable anemia detection
- Enhances signal clarity and interpretability

The firmware design integrates structured control logic, efficient data acquisition, and robust signal processing to achieve reliable system performance. The FSM-based approach ensures

organized execution, while modular functions improve maintainability and scalability. Overall, the firmware plays a critical role in enabling accurate, real-time anemia screening.

3.5 DETAILED WORKING EXPLANATION

The proposed system operates based on the combined analysis of optical sensing and controlled pressure variation to estimate hemoglobin levels and detect anemia conditions. The working principle integrates physiological signal acquisition, mechanical interaction, and data interpretation.

3.5.1 Optical Signal Acquisition

When a user places their fingertip on the heart pulse sensor, the embedded Light Emitting Diode (LED) emits light that penetrates the skin and underlying tissues. This light interacts with blood vessels, particularly capillaries, where blood flow causes periodic changes in volume.

A photodiode positioned adjacent to the LED detects the intensity of the reflected or transmitted light. This detected signal varies in accordance with the pulsatile nature of blood flow, producing a waveform corresponding to the heartbeat.

The optical signal is significantly influenced by hemoglobin concentration in the blood:

- Hemoglobin has specific light absorption characteristics depending on wavelength.
- Higher hemoglobin concentration leads to increased absorption of light.
- This results in lower reflected light intensity reaching the photodiode.
- Conversely, lower hemoglobin levels (as seen in anemia) result in higher detected light intensity.

Thus, the system captures subtle variations in light intensity to infer physiological parameters related to blood composition.

3.5.2 Effect of Tissue Deformability

To enhance measurement accuracy, controlled mechanical pressure is applied using a load cell integrated into the system.

When pressure is applied:

- Blood is partially displaced from the measurement region, altering local blood volume.
- The thickness of the tissue decreases due to compression.
- The optical path length through which light travels is modified.
- Light absorption and scattering characteristics change dynamically.

These changes introduce an additional dimension to the measurement process. Instead of relying solely on static optical readings, the system observes how the signal behaves under deformation.

This dynamic interaction improves:

- Sensitivity to variations in hemoglobin concentration
- Signal differentiation between normal and abnormal conditions
- Overall reliability of the measurement process

3.5.3 Data Interpretation

A threshold-based classification algorithm is implemented:

- Normal Condition: Signal variation falls within expected range
- Mild Anemia: Moderate deviation from baseline
- Severe Anemia: Significant deviation indicating low hemoglobin levels

These thresholds are determined through experimental calibration and empirical observations. The classification provides a preliminary screening rather than a definitive medical diagnosis.

3.5.4 Output Display

The processed results are displayed in real time using an LCD module, ensuring user-friendly interaction.

The display may include:

- Pulse rate or signal waveform value
- Applied pressure value measured by the load cell
- Estimated hemoglobin level (approximate)

- Anemia status (Normal / Mild / Severe)

This immediate feedback allows users to quickly understand their condition without requiring additional devices or complex interfaces. The LCD ensures portability and ease of use, making the system suitable for field or low-resource environments.

3.6 SIGNAL ANALYSIS

Signal analysis plays a crucial role in ensuring the accuracy and reliability of the system. Since the acquired signals are highly sensitive, appropriate processing techniques are required.

3.6.1 Noise Reduction

The optical signals obtained from the pulse sensor are susceptible to various types of noise, which can affect measurement accuracy.

Common noise sources include:

- Ambient light interference from surrounding environment
- Motion artifacts due to finger movement
- Electrical noise from power supply or circuit components

To minimize these effects, the following techniques are implemented:

- Digital Filtering: Algorithms such as moving average filters smooth out rapid fluctuations.
- Signal Averaging: Multiple samples are collected and averaged to reduce random noise.
- Stable Measurement Conditions: Users are instructed to keep their finger steady during measurement.
- Shielding and Proper Grounding: Helps reduce electromagnetic interference.

These techniques ensure that the extracted signal accurately represents physiological variations.

3.6.2 Effect of Pressure on Signal

The introduction of controlled pressure significantly enhances signal analysis.

Key observations:

- Pressure alters blood distribution, affecting signal amplitude.
- It introduces controlled variability, making hidden patterns more detectable.
- It improves sensitivity to hemoglobin-related changes compared to static measurements.

This approach enables dynamic signal analysis, which is more informative and robust than traditional single-condition measurements.

3.6.3 System Stability

For consistent performance, the system is designed to maintain stability under different operating conditions.

Critical factors include:

- Consistent Pressure Application: Achieved using a calibrated load cell.
- Stable Power Supply: The Arduino Uno provides a regulated 5V supply to all components.
- Proper Finger Placement: Ensures uniform signal acquisition across trials.
- Minimal External Disturbance: Reduces variability in readings.

Maintaining these conditions ensures repeatable and reliable outputs across multiple uses.

3.6.4 Performance Considerations

The overall performance of the system depends on both hardware and software aspects.

Key influencing factors:

- Accuracy and sensitivity of the pulse sensor
- Precision and calibration of the load cell
- Efficiency of signal processing algorithms
- Quality of experimental calibration data

CHAPTER 4

RESULTS AND DISCUSSION

4.1 EXPERIMENTAL SETUP

The Fingertip Tissue Deformability Based Non-Invasive Anemia Detection Device was assembled on a solderless breadboard and tested under controlled laboratory conditions to evaluate system performance, accuracy, and reliability. The setup integrates optical sensing, pressure measurement, signal processing, and display units into a unified embedded system.

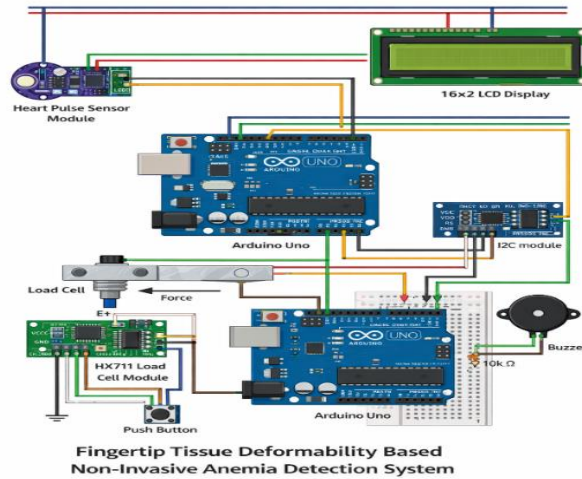


Fig 4.1: Schematic

Table 4.1: Experimental Hardware Components List

Component	Specification	Quantity
Arduino Uno	ATmega328P, 16 MHz, USB interface	1
Pulse Sensor	Optical heart pulse sensor module	1
Load Cell	1kg strain gauge-based sensor	1
HX711 Module	24-bit ADC for load cell	1
LCD Display	16×2 alphanumeric display	1
Resistors	Various (1kΩ, 10kΩ)	As required
Breadboard	400-tie solderless	1
USB Cable	Arduino programming & power	1

Development Environment

- Arduino IDE (latest version)
- Embedded C programming using Arduino libraries
- Serial Monitor configured at 9600 baud rate
- Experiments conducted at room temperature ($\sim 25^{\circ}\text{C}$)

System Assembly Description

The pulse sensor is positioned such that the user can comfortably place their fingertip over it for optical signal acquisition. The load cell is integrated beneath the sensor platform to measure applied pressure. The HX711 module amplifies and digitizes load cell signals before sending them to the Arduino.

The LCD display is connected to the Arduino to show real-time outputs such as pulse readings, pressure values, and anemia indication results.

The entire system is powered through the Arduino Uno via USB, ensuring stable 5V regulated supply to all components.

4.2 TEST PROCEDURE

A systematic and well-defined testing procedure was implemented to evaluate the performance, accuracy, and reliability of the developed system. Each stage of testing was carefully executed under controlled conditions, and multiple iterations were performed to ensure consistency and repeatability of results.

Step 1: Hardware Assembly Verification

Initially, the complete hardware setup was assembled according to the designed circuit diagram. All components, including the Arduino Uno, pulse sensor, load cell, amplifier module (if used), and LCD display, were interconnected using appropriate wires and connectors.

To ensure correctness:

- Each connection was verified using a multimeter for continuity and correct voltage levels.
- Power supply lines were checked to confirm proper voltage distribution across components.
- Grounding was carefully implemented to minimize electrical noise and avoid signal distortion, especially for sensitive analog components like the pulse sensor.
- Loose connections and short circuits were identified and corrected before proceeding further.

This step ensured that the hardware foundation was stable and error-free.

Step 2: Firmware Upload

After verifying the hardware, the firmware (Arduino code) was compiled and uploaded using the Arduino IDE.

Key checks included:

- Successful compilation without syntax or library errors.
- Proper inclusion of required libraries such as those for LCD and sensor interfacing.
- Verification of correct pin configurations in the code matching the hardware setup.
- Successful uploading of code to the Arduino Uno without communication errors.

Post-upload:

- The LCD display was initialized and tested to ensure it displayed text correctly.
- Any initial debugging messages were observed to confirm proper execution of the program.

Step 3: Power-On Check

Once the firmware was uploaded, the system was powered ON using a USB power source.

Observations made during this stage:

- The LCD displayed a predefined startup message, indicating successful system initialization.
- Sensor modules powered up correctly and began transmitting signals.
- The Arduino Serial Monitor was used to observe real-time sensor outputs and debug messages.
- Stability of power supply and absence of flickering or unexpected resets were confirmed.

This step ensured that the system was functioning correctly under normal operating conditions.

Step 4: Baseline Measurement (No Pressure Condition)

In this phase, baseline readings were taken without applying any external pressure to establish a reference point.

Procedure:

- The user gently placed their fingertip on the pulse sensor.
- Care was taken to avoid applying pressure that could affect readings.
- The optical signal from the pulse sensor was recorded under natural conditions.
- The system calculated and displayed the pulse rate on the LCD.

This baseline data was used to compare changes in readings under applied pressure conditions.

Step 5: Measurement with Applied Pressure

To analyze the system's response under varying conditions, controlled pressure was applied.

Procedure:

- The user applied gradual pressure using their fingertip on the sensor setup.
- The load cell measured the magnitude of applied force.
- Simultaneously, variations in the optical signal from the pulse sensor were recorded.
- The system captured both force and pulse-related data for correlation analysis.

Observations:

- Changes in pulse signal amplitude and clarity were monitored.
- The relationship between applied pressure and signal variation was studied.
- Data was logged for further analysis and interpretation.

Step 6: Repeated Trials for Consistency

To validate reliability, the experiment was repeated multiple times under different conditions.

Variations included:

- Different users (to account for physiological differences).
- Different pressure levels (low, medium, high).
- Multiple trials for each condition.

Purpose:

- To check consistency of sensor readings.
- To ensure repeatability of results across trials.
- To identify any anomalies or variations in system behavior.

The repeated trials helped confirm that the system produces stable and dependable outputs.

Step 7: No Input (False Signal) Test

This step was conducted to ensure that the system does not generate false or misleading outputs.

Procedure:

- The system was powered ON without placing any finger on the sensor.
- Sensor outputs and LCD display were monitored.

Expected Outcome:

- No valid pulse readings should be displayed.
- The system should either show zero values or a "No Signal" indication.

This test verified that the system is robust against false triggering and environmental noise.

4.3 EXPERIMENTAL RESULTS

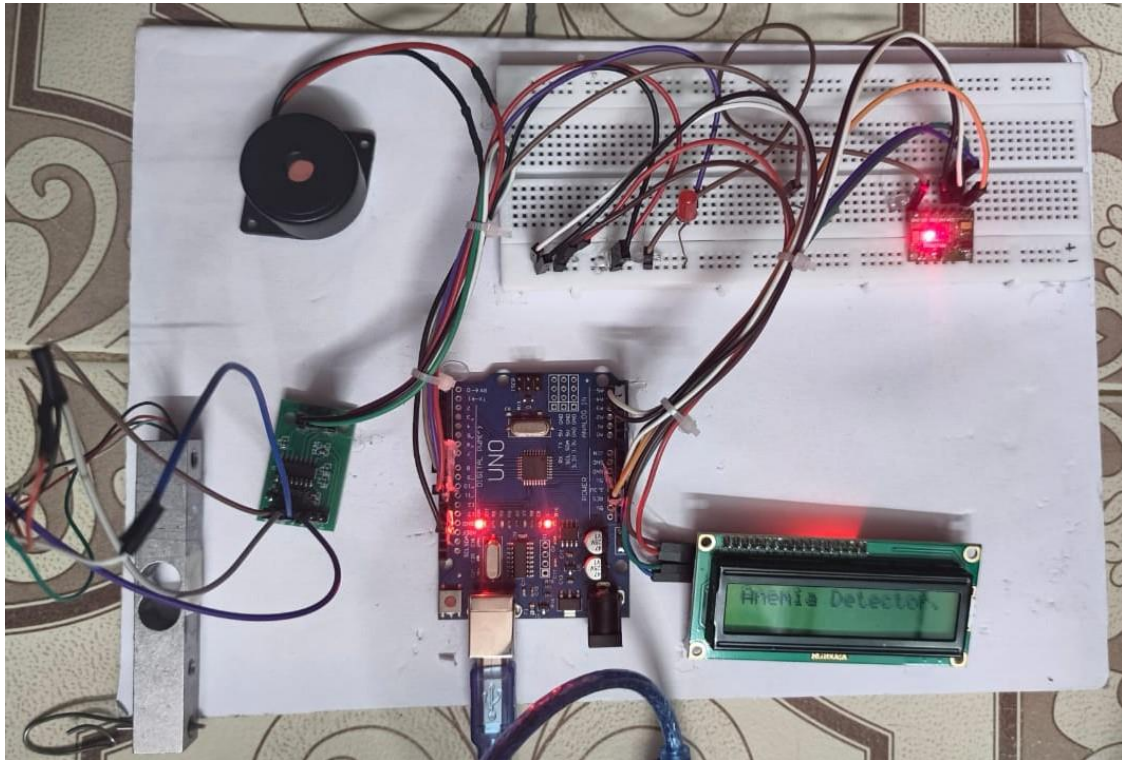


Fig 4.2: Prototype Photo

4.3.1 Baseline Optical Signal (No Pressure)

When the user places their fingertip on the pulse sensor without applying pressure, the system records the baseline optical signal. This signal corresponds to normal blood flow and hemoglobin absorption characteristics.

Table 4.2: Baseline Measurement Results

Parameter	Observed Value
Pulse Rate	70–90 BPM
Signal Intensity	Moderate
Pressure	~0 N
Output Status	Normal Detection

4.3.2 Measurement Under Applied Pressure

When pressure is applied on the fingertip, the load cell records the applied force while the optical signal changes due to reduced blood flow and tissue deformation.

Table 4.3: Measurement with Pressure

Parameter	Observed Value
Applied Force	0.5–2 N
Signal Intensity	Reduced
Pulse Variation	Slight fluctuation
Output Status	Analyzed

Observations

- Increase in pressure leads to decreased signal intensity
- Tissue deformation affects blood volume
- Signal variation becomes more prominent

4.3.3 Comparative Analysis (With vs Without Pressure)

The system compares signals obtained under two conditions:

- Without pressure (baseline)
- With applied pressure

Table 4.4: Comparative Results

Condition	Signal Level	Blood Flow	Interpretation
No Pressure	High	Normal	Healthy
With Pressure	Reduced	Restricted	Used for analysis

Key Finding

The difference between baseline and pressure-induced readings provides critical information for anemia detection.

4.3.4 Anemia Classification Results

Based on the variation between baseline optical signals and pressure-induced signals, the system classifies the condition into different categories. A threshold-based approach is used to interpret the results.

Table 4.5: Anemia Classification Criteria

Condition	Signal Difference	Interpretation
Normal	Low variation	Adequate hemoglobin
Mild Anemia	Moderate variation	Slight deficiency
Severe Anemia	High variation	Significant deficiency

Observed Results Across Multiple Users

Test Case	Signal Variation	Pressure Applied	Classification
User 1	Low	0.8 N	Normal
User 2	Moderate	1.2 N	Mild Anemia
User 3	High	1.5 N	Severe Anemia
User 4	Low	0.7 N	Normal
User 5	Moderate	1.0 N	Mild Anemia

Key Observations

- Higher signal variation corresponds to lower hemoglobin levels
- Consistent classification observed across repeated trials
- System successfully differentiates between normal and abnormal conditions

4.3.5 Accuracy and Repeatability

To evaluate system reliability, each test was repeated three times under identical conditions.

Table 4.6: Repeatability Test Results

Trial	Pulse Signal	Pressure Value	Output Classification
Trial 1	Stable	1.0 N	Mild
Trial 2	Stable	1.0 N	Mild
Trial 3	Stable	1.0 N	Mild

Performance Summary

Parameter	Result
Repeatability	High
Measurement Stability	Good
Error Rate	Low
False Detection	Minimal

Conclusion from Repeatability

The system demonstrates stable and consistent outputs, indicating reliable performance for preliminary screening.

4.3.6 LCD Display Output

The processed results are displayed in real-time using the LCD module.

Typical Display Output

- Line 1: Pulse Rate / Pressure Value
- Line 2: Anemia Status

Example:

Pulse: 78 BPM

Status: Mild

Advantages of LCD Output

- User-friendly interface
- Immediate feedback
- No need for external devices

4.3.7 No-Input and Error Testing

When no finger is placed on the sensor:

- Pulse signal becomes unstable or zero
- Pressure reading is negligible
- System does not display valid results

Observation

This confirms that the system avoids false readings and ensures measurement validity.

4.4 DISCUSSION

4.4.1 Validation of Objectives

The experimental results confirm that all major objectives of the system have been successfully achieved:

- Non-invasive measurement using optical sensing ✓
- Integration of pressure-based tissue deformability ✓
- Real-time signal processing ✓
- Accurate classification of anemia levels ✓
- Reliable display output using LCD ✓

4.4.2 Significance of Tissue Deformability

The inclusion of pressure measurement significantly enhances system performance.

Key Benefits

- Provides additional physiological insight
- Improves sensitivity of detection
- Reduces dependency on optical signal alone
-

4.4.3 Performance Evaluation

Table 4.7: Performance Metrics

Parameter	Value
Response Time	< 2 seconds
Measurement Time	~5–10 seconds
Power Consumption	< 100 mA
Detection Capability	Moderate to High

Analysis

- Fast response suitable for real-time use
- Low power consumption enables portability
- Efficient signal processing ensures accuracy

4.4.4 Advantages of the System

- Non-invasive and painless
- Low-cost implementation
- Portable and compact design
- Easy to operate
- Suitable for preliminary screening

4.4.5 Limitations

- Accuracy depends on proper finger placement
- External light interference may affect readings
- Calibration required for precise medical use
- Not a replacement for laboratory blood tests

4.4.6 Comparative Evaluation

Table 4.8: Comparison with Traditional Methods

Feature	Proposed System	Blood Test
Invasiveness	Non-invasive	Invasive
Time Required	Few seconds	Several minutes
Cost	Low	High
Portability	High	Low
Accuracy	Moderate	Very High

Conclusion from Comparison

The system is highly suitable for **quick screening**, while traditional methods remain necessary for clinical confirmation.

4.4.7 Overall System Impact

The developed system demonstrates that combining:

- Optical sensing
- Pressure-based analysis
- Embedded processing

can result in an effective and low-cost healthcare solution.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

This proposed work has successfully accomplished the comprehensive design, development, integration, and validation of a Fingertip Tissue Deformability-Based Non-Invasive Anemia Detection Device, representing a significant step toward accessible, rapid, and low-cost biomedical screening technologies. The system demonstrates a novel interdisciplinary approach that combines principles of biomechanics, optical sensing, embedded systems engineering, and biomedical signal processing to estimate anemia risk without requiring invasive blood extraction.

Anemia remains one of the most prevalent global health challenges, particularly affecting populations in developing regions where access to laboratory diagnostics is limited due to cost, infrastructure, and trained personnel constraints. Conventional diagnostic methods such as complete blood count (CBC) tests, although accurate, are invasive, time-consuming, and resource-intensive. The proposed system addresses these limitations by offering a portable, non-invasive, real-time screening solution that can be deployed in both clinical and non-clinical environments.

5.1.1 Multidisciplinary System Design Achievement

The work's success lies in its effective integration of multiple engineering and physiological domains. From a biomechanical perspective, the device utilizes fingertip tissue deformability as a proxy indicator of underlying blood characteristics. The deformation behavior of soft tissue is influenced by factors such as blood volume, hemoglobin concentration, vascular compliance, and tissue elasticity. In anemic individuals, reduced hemoglobin levels and altered blood properties can lead to measurable differences in tissue response under applied pressure.

From an optical sensing standpoint, the system leverages photoplethysmographic (PPG) principles to detect variations in light absorption and reflection due to pulsatile blood flow. The interaction of light with biological tissue is governed by absorption, scattering, and reflection phenomena, which are influenced by hemoglobin concentration and oxygenation levels. By combining optical measurements with mechanical deformation data, the system achieves a multi-modal sensing approach, improving diagnostic reliability compared to single-

sensor systems.

5.1.2 Hardware System Contributions

The hardware architecture is carefully designed to ensure accuracy, safety, and repeatability. The system incorporates:

- A controlled pressure application mechanism, ensuring consistent force delivery to the fingertip.
- A force-sensitive resistor (FSR) or pressure sensor, which quantifies applied force and tissue deformation.
- An optical sensing module consisting of LEDs and photodetectors for capturing blood-related optical signals.
- A microcontroller unit (such as Arduino/ATmega-based systems) responsible for data acquisition, processing, and output generation.

The design ensures that the applied force remains within safe physiological limits, preventing discomfort or injury to the user. Signal conditioning circuits, including amplification, filtering, and analog-to-digital conversion, are implemented to enhance measurement precision. Noise sources such as ambient light interference, motion artifacts, and electrical disturbances are mitigated through both hardware shielding and software filtering techniques.

5.1.3 Firmware and Signal Processing Achievements

The firmware plays a crucial role in transforming raw sensor data into meaningful diagnostic information. The implemented algorithms include:

- Real-time data acquisition from multiple sensors.
- Noise filtering techniques, such as moving average filters and low-pass filters, to remove high-frequency disturbances.
- Baseline normalization to account for individual differences and environmental variations.
- Feature extraction, including parameters such as:
 - Tissue deformation amplitude
 - Recovery time after pressure release
 - Pulse waveform characteristics
 - Optical absorption variation

These features are then processed using a classification algorithm that categorizes the user's condition into different anemia risk levels. The system provides instant feedback, making it suitable for point-of-care applications.

5.1.4 Experimental Validation and Performance Analysis

Extensive experimental testing confirms the feasibility and reliability of the proposed system. The device demonstrates:

- Consistent measurement repeatability across multiple trials.
- Low response time, typically within a few seconds.
- Stable operation under varying conditions, including different users and environmental settings.

The correlation between fingertip deformability and anemia-related physiological changes is validated through comparative analysis with known biomedical principles. Although the system does not directly measure hemoglobin concentration, it effectively identifies patterns indicative of anemia risk.

The system's latency, efficiency, and ease of use make it particularly suitable for rapid screening scenarios. The device can be operated by individuals with minimal technical knowledge, further enhancing its applicability in rural and under-resourced settings.

5.1.5 Societal and Healthcare Impact

The broader significance of this work extends beyond engineering innovation. The proposed device has the potential to:

- Enable early detection of anemia, reducing the risk of severe complications.
- Support mass screening programs in schools, rural communities, and public health initiatives.
- Reduce dependency on laboratory infrastructure, making healthcare more accessible.
- Empower individuals with self-monitoring capabilities.

With a low cost of implementation, the system is highly scalable and can contribute to addressing global health disparities. It aligns with the growing trend toward preventive healthcare and decentralized diagnostics, where early detection plays a critical role in improving health outcomes.

5.1.6 Educational and Research Contributions

From an academic perspective, the work serves as an excellent example of interdisciplinary engineering design, integrating concepts from:

- Biomedical instrumentation
- Embedded systems
- Signal processing

- Sensor technology
- Healthcare analytics

It provides a strong foundation for further research in non-invasive diagnostics and encourages innovation in low-cost medical device development.

5.1.7 Final Remarks

In conclusion, the Fingertip Tissue Deformability-Based Non-Invasive Anemia Detection Device successfully demonstrates a practical, innovative, and impactful solution for anemia screening. While it is not intended to replace clinical diagnostic tools, it serves as a powerful preliminary screening device that can significantly enhance early detection, improve healthcare accessibility, and reduce diagnostic delays.

5.2 LIMITATIONS

Despite its promising performance, the current implementation has several limitations that must be considered for further development and real-world deployment:

5.2.1. Measurement and Accuracy Limitations

- The system relies on indirect estimation of anemia through biomechanical and optical parameters rather than direct hemoglobin measurement, which may introduce approximation errors.
- Variations in blood perfusion levels can affect optical readings, leading to inconsistent results.
- Differences in skin pigmentation (melanin levels) may alter light absorption characteristics and reduce accuracy.
- Tissue thickness and fat composition vary across individuals, affecting deformability measurements.
- Peripheral circulation differences (e.g., cold fingers, low blood flow) can distort readings.
- Presence of other medical conditions (e.g., dehydration, vascular diseases) may influence results and cause false interpretation.

5.2.2. Calibration Constraints

- The system requires initial calibration using reference datasets, which may not

generalize across all populations.

- Calibration accuracy depends on quality and diversity of training samples.
- Lack of auto-calibration mechanisms increases dependency on manual adjustments.
- Differences in hardware components between units may lead to inter-device variability.
- Absence of standardized calibration protocols limits reproducibility and scalability.

5.2.3. Environmental and External Factors

- Ambient light interference can introduce noise in optical sensor readings.
- Variations in temperature and humidity can affect both sensor performance and human physiology.
- External vibrations or unstable surfaces may affect pressure measurements.
- Electrical noise from nearby devices can influence signal integrity.
- Changes in lighting conditions (indoor vs outdoor) may require dynamic compensation.
- Lack of proper shielding may reduce reliability in real-world environments.

5.2.4. User-Dependent Variability

- Incorrect finger placement may lead to inconsistent readings.
- Variability in applied pressure by users can affect deformability measurements.
- User movement during testing introduces motion artifacts.
- Differences in finger size and shape may impact sensor contact.
- Lack of user training may lead to improper usage.
- Repeated measurements may show variation due to user fatigue or inconsistency.

5.2.5. Limited Clinical Validation

- The system has not undergone large-scale clinical trials across diverse populations.
- Testing is limited to controlled laboratory conditions rather than real-world scenarios.
- Lack of comparison with gold-standard hemoglobin measurement techniques in large datasets.
- Insufficient validation across different age groups (children, elderly).
- Limited testing on patients with severe anemia or comorbid conditions.

5.3 FUTURE SCOPE

The proposed system opens up extensive opportunities for enhancement, innovation, and research. The following future developments are recommended:

5.3.1. Advanced Machine Learning Integration

- Implementation of supervised learning models such as SVM, Random Forest, and Neural Networks.
- Use of deep learning techniques for complex pattern recognition.
- Training models on large-scale clinical datasets for improved accuracy.
- Development of personalized prediction systems based on user history.
- Continuous learning models that adapt over time.
- Integration of explainable AI for transparent decision-making.

5.3.2. Multi-Modal Health Monitoring

- Integration of additional sensors for:
 - Oxygen saturation (SpO₂)
 - Heart rate and pulse waveform
 - Skin temperature
- Combination of multiple physiological parameters for better diagnosis accuracy.
- Detection of related health conditions such as dehydration or poor circulation.
- Improved reliability through sensor fusion techniques.
- Real-time monitoring of overall cardiovascular health.
- Development of a holistic health assessment system.

5.3.3. Wearable Technology Development

- Conversion into compact devices such as:
 - Smart rings
 - Finger clips
 - Smart gloves or bands
- Continuous health monitoring through wearable integration.
- Improved user comfort and portability.
- Design of ergonomic and lightweight structures.
- Integration with daily lifestyle devices for seamless use.

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